SALESFORECAST

CSV-Based Sales Analytics & Forecasting Tool

Project Report

Submitted by

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CERTIFICATE

This is to certify that the mini project entitled

SALESFORECAST — CSV-BASED SALES ANALYTICS & FORECASTING TOOL

has been successfully developed and submitted by **NIKHIL** (Roll No. **24BCA088**), a student of the 4th Semester, Bachelor of Computer Applications (BCA), St. Claret College (Autonomous), Bengaluru.

This mini project was developed independently as part of self-learning during the vacation period. The work is original and has not been submitted elsewhere for any academic examination or award.

St. Claret College (Autonomous), Bengaluru

DECLARATION

I, **NIKHIL**, Roll No. **24BCA088**, 4th Semester, BCA, St. Claret College (Autonomous), Bengaluru, declare that the mini project **"SALESFORECAST"** is a record of original work done by me during the vacation period.

I declare that:

- This project was carried out independently, without any formal institutional guidance.
- The work is genuine and has not been copied from any source without acknowledgement.
- This project has not been submitted for any degree or academic title elsewhere.
- All references have been duly cited.
- The conclusions drawn are based entirely on my own understanding and research.

Place: Bengaluru

Date: July 2026

Signature of the Student

NIKHIL

Roll No: 24BCA088 | 4th Semester, BCA

ACKNOWLEDGEMENT

I thank all those who encouraged and supported me throughout the development of this project.

I am grateful to the open-source community and educational resources available through MDN Web Docs, Stack Overflow, and YouTube, which helped me understand CSV parsing techniques, JavaScript-based linear trend forecasting, Chart.js visualisation, and drag-and-drop file handling in the browser.

I thank my family for their constant encouragement and for providing a focused environment for learning during the vacation period.

Building a tool that takes raw business data and transforms it into visual insights and predictive forecasts — entirely in the browser, with no server — gave me a deep appreciation of how software can solve real-world business problems using only front-end technologies.

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ABSTRACT

Small and medium-sized businesses frequently struggle with making data-driven decisions due to the complexity and cost of professional analytics tools. Sales data is often maintained in spreadsheets but never analysed systematically — leaving business owners without visibility into their top-performing products, revenue trends, or future projections. SalesForecast addresses this gap by offering a free, browser-based sales analytics and forecasting tool that requires nothing more than a CSV export from Excel or Google Sheets.

SalesForecast is a three-step web application — Upload, Analyse, Forecast — that reads a user's sales CSV file, parses it entirely in the browser, and instantly generates a comprehensive analytics dashboard including revenue charts, product comparisons, a data preview table, key business insights, and a 3-month forward sales forecast. No data is sent to any server at any point.

Key Features:

- Drag-and-drop or browse CSV file upload with automatic column detection (date, product, quantity, revenue).
- Overview metrics: Total Revenue, Total Units Sold, Best Product, and Date Range.
- Revenue Over Time: a bar chart plotting monthly revenue from the uploaded data.
- Units Sold Over Time: a bar chart tracking monthly unit volume.
- Revenue by Product: a ranked horizontal bar chart with colour-coded product comparison.
- Sales Data Preview: a paginated table showing the first 20 rows of uploaded data.
- 3-Month Sales Forecast: a projected revenue estimate using average monthly growth rate.
- Key Insights: auto-generated business tips — Best Month, Top Product, Lowest Product.
- Fully client-side — no server, no account, no data transmission.

■■ IMPORTANT NOTE FOR USERS For the Sales Forecast feature to generate meaningful and accurate predictions, the uploaded CSV file MUST contain sales data spanning a minimum of TWO (2) MONTHS. The forecasting algorithm relies on identifying monthly growth trends across multiple data points. If only a single month of data is provided, the application will still display analysis results, but the 3-month forecast projection will be based solely on an average growth rate and may not reflect actual future sales patterns accurately. Ensure your CSV includes date, product, quantity/sales, and revenue/amount columns covering at least two distinct calendar months before uploading.

Technologies Used:

HTML5, CSS3, Vanilla JavaScript, Chart.js, FileReader API. Deployed on Netlify.

TABLE OF CONTENTS

Certificate	2
Declaration	3
Acknowledgement	4
Abstract	5
Table of Contents	6
Chapter 1 – Introduction	7
Chapter 2 – Objectives of the Project	8
Chapter 3 – Existing System	9
Chapter 4 – Proposed System	10
Chapter 5 – Modules and Features	11
Chapter 6 – Technologies Used	15
Chapter 7 – Conclusion	17

Chapter 1

INTRODUCTION

1.1 Background

Sales forecasting is one of the most fundamental activities in business management. Knowing which products are performing well, which months generate the most revenue, and what the next quarter might look like allows businesses to manage stock, allocate marketing budgets, and plan expansions effectively. However, most forecasting tools are either enterprise-grade software requiring installation and licences, or online SaaS platforms that require account creation and charge monthly fees.

Small retailers, student entrepreneurs, and freelancers who maintain their sales records in Excel or Google Sheets need a simpler, immediate solution — one that accepts the data they already have and returns actionable insights without friction.

1.2 About the Project

SalesForecast is a lightweight, browser-based sales analytics tool that accepts any well-structured CSV file and instantly produces a full analytics dashboard and 3-month sales forecast. The three-step workflow — Upload → Analyse → Forecast — is intuitive enough for non-technical users while being technically sophisticated in its CSV parsing, chart rendering, and growth-rate projection logic.

Live at: **https://sales-forcast.netlify.app/** — deployed on Netlify.

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1.3 Scope

- CSV file upload with automatic detection of date, product, quantity, and revenue columns.
- Revenue and unit volume charting by month and by product.
- 3-month forward sales forecast using average monthly growth rate.
- Auto-generated business insights based on the uploaded data.

- Fully client-side — no server, no account, no data storage beyond the session.

OBJECTIVES OF THE PROJECT

The following objectives guided the design and development of SalesForecast:

1. To build a browser-based CSV analytics engine: Parse uploaded CSV files entirely in JavaScript using the FileReader API, automatically identifying relevant columns (date, product, quantity, revenue) regardless of exact column names.

2. To visualise sales performance through interactive charts: Render bar charts for monthly revenue and unit volume, and a ranked horizontal bar chart for product-wise revenue comparison, using Chart.js with a clean, consistent colour palette.

3. To implement a 3-month sales forecast: Calculate an average monthly growth rate from historical data and project revenue for the next three months, displaying the forecast with an appropriate uncertainty disclaimer.

4. To generate automated business insights: Identify and highlight the best month, top product, and lowest-performing product automatically from the uploaded data, providing immediately actionable guidance.

5. To provide a transparent data preview: Display the first 20 rows of the uploaded file in a formatted table so users can verify that their data was read correctly before trusting the analysis.

6. To ensure a friction-free user experience: Support both drag-and-drop and browse-file upload, provide a clear 3-step progress indicator, and include a CSV format guide so users know exactly what data to prepare.

7. To maintain complete data privacy: Process all data locally in the browser — no file contents are transmitted to any server, ensuring confidentiality of business data at all times.

EXISTING SYSTEM

3.1 Current Approaches to Sales Analysis

Businesses and students currently rely on the following methods for sales data analysis, each with significant limitations:

a) Manual Spreadsheet Analysis

Most small businesses maintain sales data in Microsoft Excel or Google Sheets and analyse it manually using SUM/AVERAGE formulas and basic charts. This process is time-consuming, error-prone, and produces no forward-looking forecasts without complex formula construction.

b) Enterprise BI Tools

Tools like Tableau, Power BI, or Looker Studio offer powerful analytics but require installation or account creation, carry licensing costs, and have steep learning curves that make them impractical for students or small retailers.

c) Online Analytics SaaS Platforms

Cloud-based tools like Zoho Analytics or Freshsales require account registration and subscription fees. Uploading sensitive business data to a third-party server also raises privacy concerns for small business owners.

d) No Forecasting in Basic Tools

Excel can produce trend lines but requires manual chart configuration. No consumer-grade free tool combines CSV upload, instant analysis, and 3-month forecasting in a single, browser-based interface.

3.2 Limitations Summary

- Manual analysis is slow, error-prone, and produces no forecasts.
- Enterprise tools require licensing, installation, and training.
- Cloud tools involve uploading sensitive business data to third-party servers.
- No free, instant, browser-based tool combines upload + analysis + forecast in one workflow.
- Existing tools are not designed for non-technical users.

PROPOSED SYSTEM

4.1 Overview

SalesForecast proposes a zero-friction, browser-based sales analytics platform that takes a standard CSV export from any spreadsheet tool and instantly produces a complete analytics dashboard and 3-month forecast — privately, in the browser, without any account or server dependency.

4.2 Advantages

- **Instant Analysis:** Upload a CSV and receive a full analytics dashboard — charts, insights, forecast — within seconds.
- **Zero Setup:** No installation, account, or subscription required. Open the URL and upload.
- **Data Privacy:** All processing occurs in the browser. Business sales data is never transmitted to any server.
- **3-Month Forecast:** Growth rate projection gives businesses a forward-looking view that spreadsheet formulas typically do not.
- **Automated Insights:** Best Month, Top Product, and Lowest Product are identified automatically without the user writing any formula.
- **Flexible CSV Support:** Automatic column detection handles varying column names (sales/quantity, revenue/amount) without requiring a fixed template.

4.3 Architecture

SalesForecast is a single-page application with a linear three-step workflow. Step 1 (Upload) uses the HTML5 FileReader API to read the CSV file into memory. Step 2 (Analyse) parses the CSV using a custom JavaScript parser, groups data by month and product, and feeds the results to Chart.js for rendering. Step 3 (Forecast) applies the average monthly growth rate formula to project the next three months' revenue. All state exists only in memory during the browser session.

Chapter 5

MODULES AND FEATURES

SalesForecast is structured as a three-step linear workflow with the following modules.

5.1 Step 1 — Upload Module

The Upload module is the entry point of the application. A large drag-and-drop zone invites users to upload their CSV file. A 'Browse File' button provides an alternative for users who prefer a file dialog. Upon file selection, the application confirms the upload with a green success banner showing the filename, row count, and column count detected.

A collapsible 'What should my CSV look like?' section explains the required column structure with a reference table covering the date, product, sales/quantity, and revenue/amount fields along with example values. A tip informs users that they can export directly from Excel or Google Sheets.

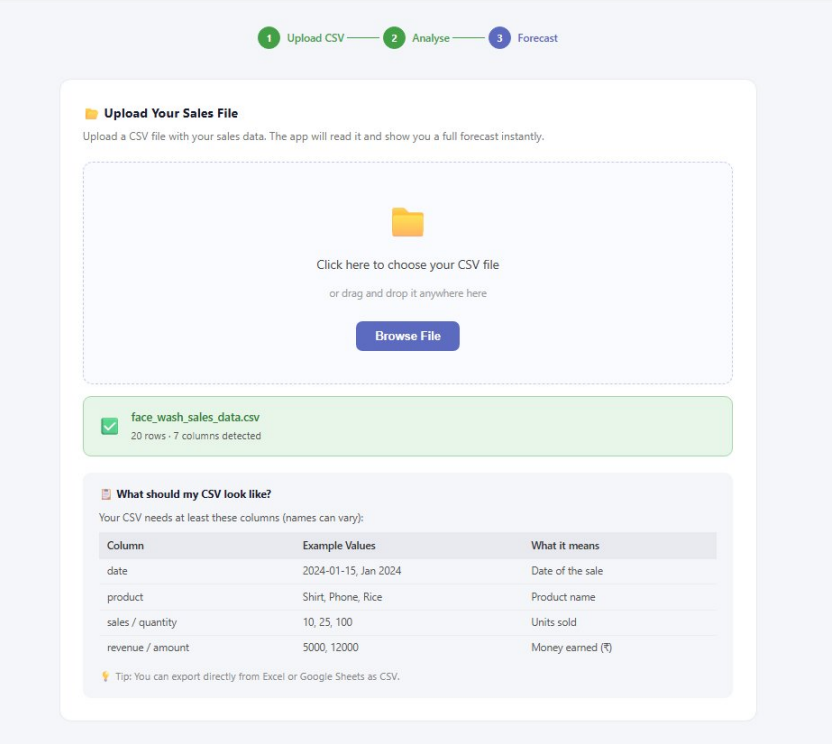


Figure 5.1 – Upload Module with Drag-and-Drop Zone and CSV Format Guide

5.2 Step 2 — Analysis: Overview and Revenue Chart

After upload, the application transitions to the analysis view displaying four overview metric cards:

- **Total Revenue:** Cumulative revenue across all rows in the dataset.
- **Total Units Sold:** Cumulative quantity across all transactions.
- **Best Product:** The product with the highest total revenue, with amount earned.
- **Date Range:** The earliest and latest dates present in the uploaded data.

Below the metrics, a Revenue Over Time bar chart displays total monthly revenue. Each bar represents one calendar month's total earnings, allowing users to identify seasonal peaks and troughs at a glance.

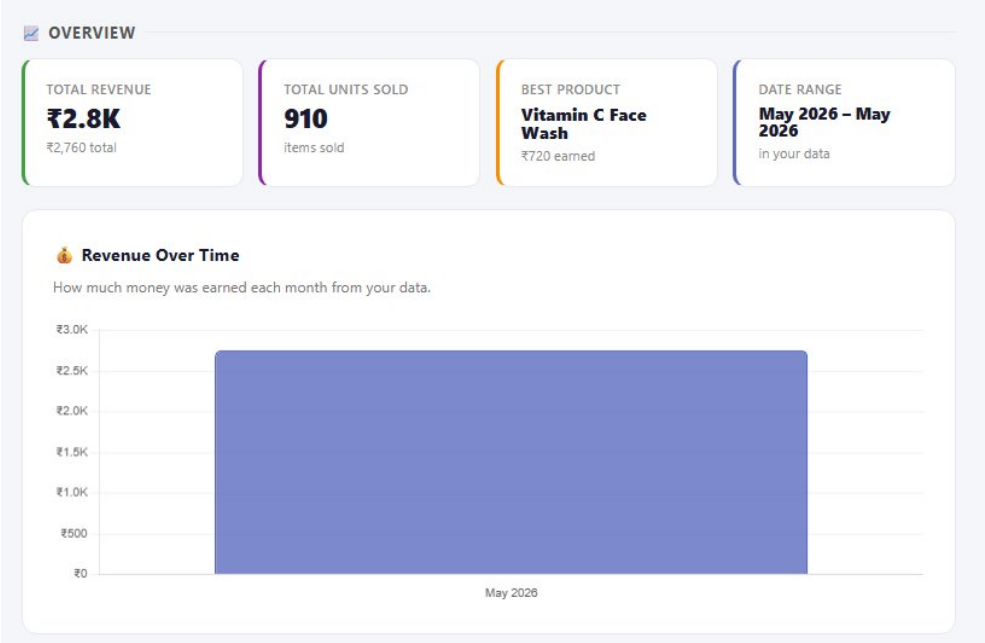


Figure 5.2 – Overview Metrics and Revenue Over Time Chart

5.3 Revenue by Product

The Revenue by Product section displays a ranked list of all products in the dataset, sorted by total revenue. Each product row shows a colour-coded horizontal progress bar, the total revenue earned, and the total units sold. This gives users an immediate comparative view of product performance across the entire sales period.

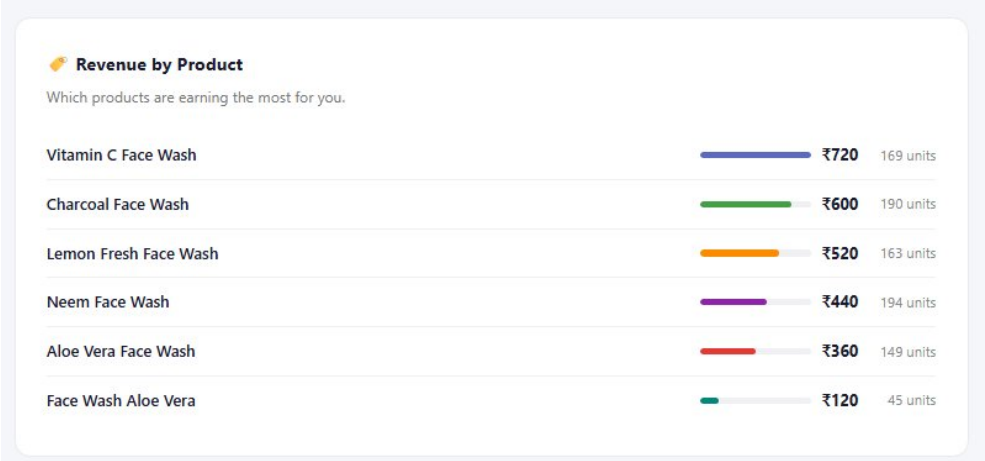


Figure 5.3 – Revenue by Product with Ranked Horizontal Bar Indicators

5.4 Sales Data Preview Table

The Sales Data Preview section renders the first 20 rows of the uploaded CSV in a clean, formatted table with column headers corresponding to the detected fields. This allows users to verify that the application correctly parsed their data — including date formatting, product names, unit values, and revenue figures — before trusting the analysis and forecast outputs.



Your Sales Data (Preview)

First 20 rows from your uploaded file.

date	product	region	units sold	unit price	total sales	salesperson
2026-05-01	Face Wash Aloe Vera	Bangalore	45	120	5400	Rahul
2026-05-02	Charcoal Face Wash	Chennai	38	150	5700	Sneha
2026-05-03	Vitamin C Face Wash	Hyderabad	52	180	9360	Arjun
2026-05-04	Neem Face Wash	Mumbai	40	110	4400	Priya
2026-05-05	Lemon Fresh Face Wash	Delhi	35	130	4550	Karan
2026-05-06	Aloe Vera Face Wash	Pune	48	120	5760	Meera
2026-05-07	Charcoal Face Wash	Bangalore	55	150	8250	Vikram
2026-05-08	Vitamin C Face Wash	Chennai	42	180	7560	Anjali
2026-05-09	Neem Face Wash	Hyderabad	50	110	5500	Rohit
2026-05-10	Lemon Fresh Face Wash	Mumbai	30	130	3900	Kavya
2026-05-11	Aloe Vera Face Wash	Delhi	60	120	7200	Rahul
2026-05-12	Charcoal Face Wash	Pune	44	150	6600	Sneha
2026-05-13	Vitamin C Face Wash	Bangalore	36	180	6480	Arjun
2026-05-14	Neem Face Wash	Chennai	58	110	6380	Priya
2026-05-15	Lemon Fresh Face Wash	Hyderabad	47	130	6110	Karan
2026-05-16	Aloe Vera Face Wash	Mumbai	41	120	4920	Meera
2026-05-17	Charcoal Face Wash	Delhi	53	150	7950	Vikram
2026-05-18	Vitamin C Face Wash	Pune	39	180	7020	Anjali
2026-05-19	Neem Face Wash	Bangalore	46	110	5060	Rohit
2026-05-20	Lemon Fresh Face Wash	Chennai	51	130	6630	Kavya

Figure 5.4 – Sales Data Preview Table Showing First 20 Rows of Uploaded CSV

5.5 Step 3 — Forecast and Key Insights

3-Month Sales Forecast:

The forecast module calculates the average month-on-month revenue growth rate from the historical data and projects this rate forward for three months. The projected values are displayed with a clear disclaimer that the estimate is based on average growth and actual results may vary.

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Key Insights:

Three auto-generated insight cards appear below the forecast:

- **Best Month:** Identifies the calendar month with the highest revenue and states the exact amount earned.
- **Top Product:** Names the highest-revenue product and its total earnings, with a recommendation to keep it stocked.
- **Lowest Product:** Flags the lowest-revenue product and suggests reviewing pricing or promotion strategies.

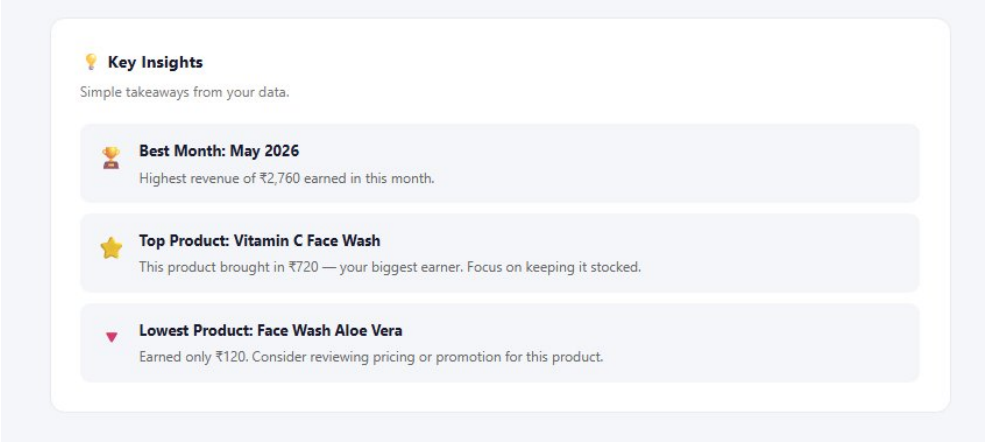


Figure 5.5 – Key Insights Module with Best Month, Top Product, and Lowest Product

TECHNOLOGIES USED

SalesForecast was developed using only native front-end web technologies. No back-end server or database is used.

6.1 HTML5

HTML5 provides the structural foundation of the three-step workflow layout. The File API (specifically FileReader) is the core enabling technology — it reads the user's CSV file into memory as a text string without any server upload. A drag-and-drop zone is implemented using HTML5 drag events. The progress step indicators use HTML5 semantic structure.

6.2 CSS3

CSS3 delivers the application's clean, light-themed interface with a pale grey background and white card surfaces. CSS Custom Properties manage the colour system, with green accents for success states and purple for chart colours. CSS transitions animate the step indicator progression and the success banner appearance on file upload. Flexbox and CSS Grid handle the responsive layout of the metric cards and product list.

6.3 JavaScript (ES6+, Vanilla)

All application logic is implemented in Vanilla JavaScript. Key algorithms include: a custom CSV parser that splits by line and handles quoted fields, a column auto-detection function that maps column names to their semantic role (handling synonyms like 'sales'/'quantity' and 'revenue'/'amount'), a date parsing and grouping function that aggregates rows by calendar month, a product aggregation function that totals revenue and units per product, and an average growth rate forecasting function projecting three months ahead.

6.4 Chart.js

Chart.js renders three chart types: a bar chart for revenue over time, a bar chart for units sold over time, and a horizontal bar visualisation for the product revenue comparison. Charts are initialised with the grouped data after CSV parsing completes and update dynamically if a new file is uploaded.

6.5 FileReader API

The HTML5 FileReader API is the critical browser API that enables SalesForecast's privacy-first architecture. It reads the selected or dropped CSV file as text directly in the browser, making the contents available to JavaScript without any network request. This ensures that sensitive business data never leaves the user's device.

6.6 Netlify

The application is deployed as a static site on Netlify, providing HTTPS, a custom subdomain, and global CDN distribution with no server-side infrastructure.

#	Technology	Type	Purpose
1	HTML5	Markup	Structure, drag-and-drop, File API
2	CSS3	Styling	Light theme, card layout, step indicators
3	JavaScript ES6+	Programming	CSV parsing, analysis, forecasting
4	Chart.js	Library	Revenue, units, and product bar charts
5	FileReader API	Browser API	Client-side CSV file reading
6	Netlify	Platform	Static hosting and CDN deployment

CONCLUSION

SalesForecast successfully bridges the gap between raw spreadsheet data and actionable business intelligence for non-technical users. By accepting a standard CSV export — the format already used by millions of small business owners and students — and returning a complete analytics dashboard and 3-month forecast in seconds, the application delivers genuine business value with zero technical friction.

The technical architecture reflects a principled design philosophy: all processing is client-side, all data remains private, and the tool is accessible to anyone with a browser and a CSV file. Implementing a custom CSV parser, an automatic column detection system, a monthly grouping algorithm, and a growth-rate forecasting model in Vanilla JavaScript demonstrated the expressive power of the core web platform.

The Key Insights module — which automatically identifies the best month, top product, and lowest performer — is particularly noteworthy as it translates raw data into named, narrative business guidance. This is exactly the kind of feature that distinguishes a useful tool from a mere data display.

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In conclusion, SalesForecast is a focused, well-designed, and publicly deployed sales analytics tool that fully achieves its stated objectives and demonstrates a clear understanding of both front-end development techniques and the real-world problem it was built to solve.

Live Project: https://sales-forcast.netlify.app/